

ATOMQUEST HACKATHON 1.0

Syncora

Enterprise Goal Setting & Tracking Portal

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2 Project Overview

Modern enterprises face a critical challenge: executive strategy is rarely reflected in daily execution. Traditional goal management relies on static annual reviews, fragmented spreadsheets, and siloed communication, leading to misaligned priorities and delayed visibility into operational bottlenecks.

Syncora is a premium, cloud-native enterprise goal setting and tracking portal engineered to bridge this gap. Designed with a zero-trust architecture, Syncora replaces annual guesswork with dynamic **quarterly reviews**, automated Service Level Agreement (SLA) escalations, and real-time telemetry.

Core Enterprise Pillars

- **Organizational Alignment:** Horizontally and vertically cascades high-level corporate thrust areas down to individual employee Key Performance Indicators (KPIs).
- **Quarterly Reviews:** Structured milestone check-ins tracking planned vs. actual achievements with automated progress calculations.
- **RBAC Workflows:** Fine-grained, 4-tier role governance (Employee, Manager, Admin, Super Admin) ensuring strict multi-tenant data isolation.
- **Real-Time Visibility:** WebSocket-driven Change Data Capture (CDC) instantly reconciles client state without expensive polling.
- **Audit Trails:** Immutable, tamper-evident event logging tracking every entity mutation, access request, and approval lifecycle.
- **Cloud-Native Architecture:** Fully decoupled, serverless edge deployment built for five-nines (99.999%) reliability and elastic scalability.

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Core Features

Syncora delivers a robust suite of enterprise capabilities designed to automate governance and streamline goal alignment across the organization.

Goal Creation & Approval Workflow

Multi-tier submission engine with automated weightage validation, draft persistence, and inline manager comment dialogs.

Quarterly Achievement Tracking

Milestone-driven check-in panels calculating planned vs. actual achievement with real-time progress status bars.

Shared Goals

Cross-departmental KPI synchronization enabling horizontal collaboration across disparate teams and business units.

Validation Engine

Strict mathematical enforcement of 100% cumulative weightages (Sum = 100%), max 8 goals, and min 10% per goal.

Audit Logs

Cryptographic-grade event streams capturing every entity mutation, state transition, and administrative override.

Reporting Dashboards

Executive charts and telemetry displays visualizing organizational velocity, department health, and thrust area distribution.

Role-Based Access Control

Zero-trust 4-tier hierarchy enforcing strict tenant data partitioning across UI shells, API routes, and database mutations.

Analytics & Insights

Advanced AI-powered intelligence mesh delivering QoQ performance trajectories, predictive bottlenecks, and risk alerts.

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Bonus Features

Going beyond standard CRUD functionality, Syncora integrates deeply into the enterprise software ecosystem to ensure seamless adoption and high operational efficiency.

Microsoft Teams Integration

Dispatches rich Adaptive Cards (v1.4) directly to Teams channels for instant workflow actionability (e.g., direct approval/rejection from chat).

Azure AD / OAuth Authentication

Enterprise Single Sign-On (SSO) backed by NextAuth OIDC, Azure Entra ID, and secure corporate identity verification.

Redis Caching

Upstash Redis rate limiting and TanStack Query cache reconciliation ensuring sub-millisecond API response times.

CI/CD Pipeline

Automated GitHub Actions quality gates enforcing strict linting, TypeScript type safety, and zero-downtime Vercel deployments.

Enterprise UI/UX

Clean, executive aesthetic featuring clear typography, structured data tables, and calm spring motion architecture.

Automated Testing

Comprehensive STLC suite covering unit tests, integration flows, and RBAC boundary validation.

Escalation Workflows

Inngest background daemons monitoring check-in thresholds (<70%) and triggering automated multi-level SLA escalations (Manager → HR → Executive).

5 Architecture Section

Syncora operates on a fully decoupled, multi-layered enterprise architecture designed for maximum reliability, multi-tenant data isolation, and real-time responsiveness. Featuring serverless edge rendering on Vercel, the platform ensures cloud-native scalability, strict RBAC security, and immutable audit logging.

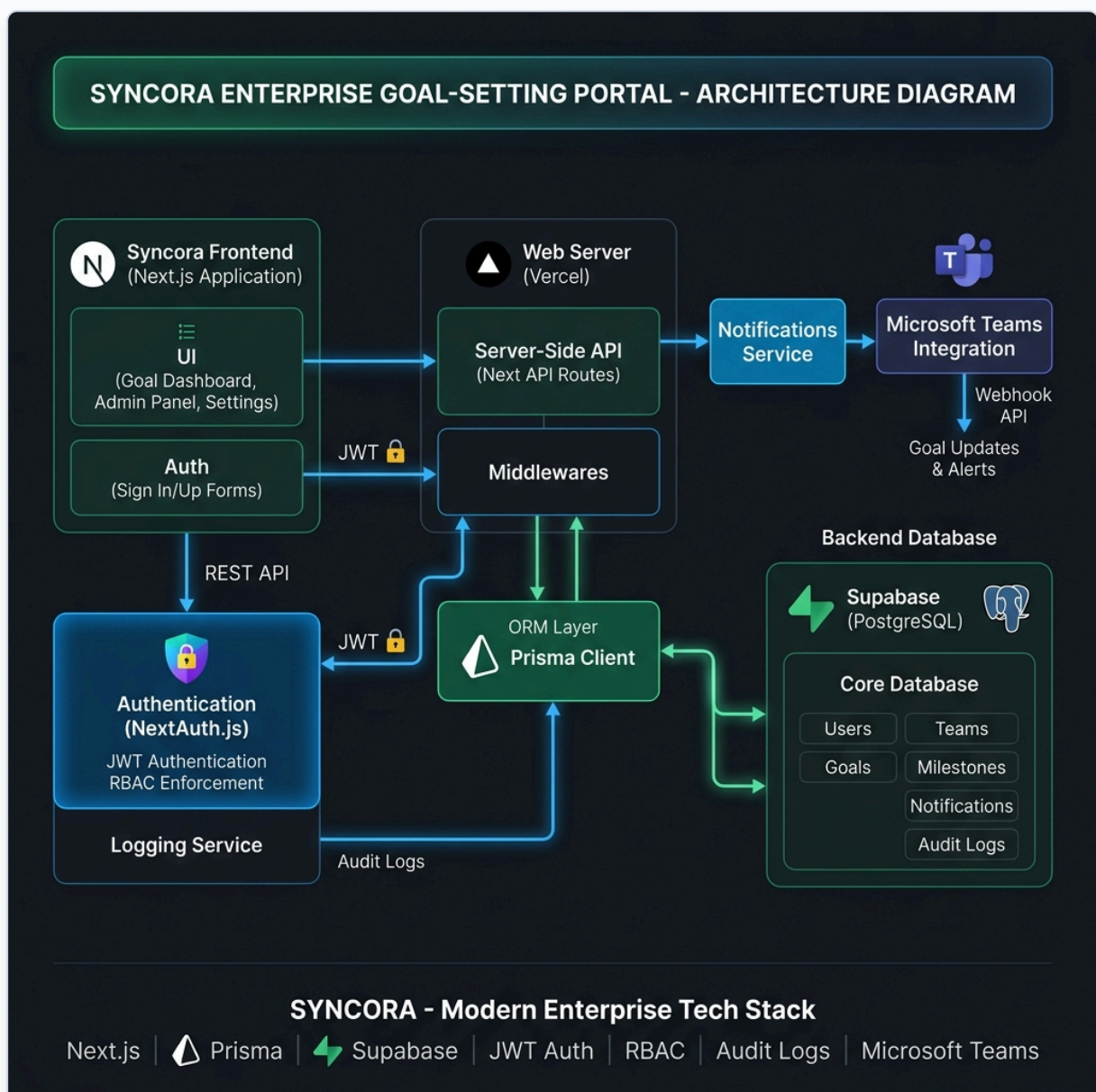


Figure 1: Cloud-Native Enterprise System Architecture

The platform is built on a robust, modern engineering foundation optimized for enterprise performance and developer velocity.

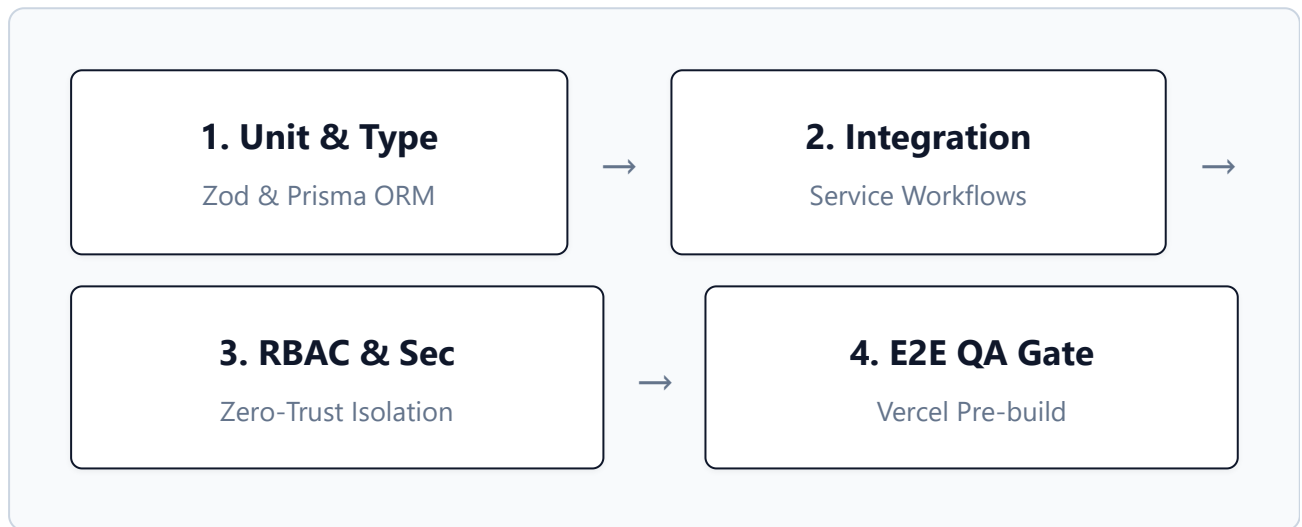
Technology	Layer / Role	Enterprise Purpose & Implementation
Next.js 16.2	Frontend & API Gateway	App Router, Server Actions, and Edge Middleware for optimized SSR/ISR.
React 19	UI Library & 3D Canvas	Component orchestration and React Three Fiber (R3F) interactive Goal Galaxy.
Tailwind CSS 4	Styling System	Utility-first design tokens, clean spacing utilities, and responsive layouts.
Prisma 6	ORM Layer	Type-safe database schema migrations, client generation, and relation pooling.
Supabase PostgreSQL	Database & CDC	Primary relational persistence, pgvector AI embeddings, and Realtime WebSockets.
NextAuth.js	Security & Identity	Enterprise OIDC / Azure Entra ID SSO authentication and secure JWT management.
Upstash Redis	Cache & Rate Limiting	Distributed in-memory caching and sliding-window API rate limiting.

Technology	Layer / Role	Enterprise Purpose & Implementation
Microsoft Teams	Enterprise Webhooks	Webhook-driven Adaptive Cards (v1.4) for actionable chat-based approvals.
GitHub Actions	CI/CD Pipeline	Automated quality gates, type checking, linting, and continuous deployment.
Vercel	Cloud Production Host	Zero-downtime serverless deployment, edge functions, and global telemetry.

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STLC & Testing Methodology

Syncora enforces a rigorous Software Testing Life Cycle (STLC) ensuring absolute reliability across all layers. Our automated CI/CD QA pipeline validates Unit testing, Integration testing, RBAC validation, Security validation, and end-to-end Workflow testing before every production release.



Testing Coverage Breakdown

- **Unit Testing:** Comprehensive test coverage of domain business logic, mathematical weightage calculations (Sum = 100%), and Zod schema parsers.
- **Integration Testing:** End-to-end verification of service boundaries, database transaction rollbacks, and Redis caching layers.
- **RBAC Validation:** Automated assertion of tenant isolation boundaries, verifying that Employee, Manager, and Admin tokens cannot access unauthorized tenant IDs.
- **CI/CD QA Pipeline:** Mandatory GitHub Actions workflows blocking pull requests if TypeScript compilation, ESLint rules, or unit tests fail.
- **Security Validation:** HMAC webhook signature verification, SQL injection prevention via Prisma ORM, and Redis rate limiter boundary testing.
- **Workflow Testing:** Simulated lifecycle execution of goal drafting, manager approval/rejection, quarterly check-in submissions, and Inngest SLA escalation

triggers.

Deployment & Access Credentials

Syncora is fully deployed and ready for live judging. Explore the live environment using the pre-seeded enterprise credentials below.

Live Hosted Demo URL: <https://syncora-3mmstahnc-hardikbhalekar10-6644s-projects.vercel.app/>

GitHub Repository: <https://github.com/hardik-bhalekar/Syncora>

Seeded Demo Credentials (Demo@123)



Employee

employee@syncora.com
Demo@123

- 1 Approved Goal Sheet
- 1 Completed Q3 Check-in
- 1 Assigned Shared Goal



Manager

manager@syncora.com
Demo@123

- Direct Manager for Employee
- Owner of Shared Enterprise Goal
- Pending Approval Queue



Admin

admin@syncora.com
Demo@123

- Full Tenant Admin Access
- Immutable Audit Log Streams
- Goal Sheet Unlocking Capabilities